Exam #2 ECEN 470 Review Chp 4 and 5

1. Know how to determine system type for a system with unity feedback.
2. Know the transfer function (form and number of poles and zeros) for a P, PI, PD, and PID compensator and how they affect system type and system dynamics.
3. Understand the steady-state error corresponding to system type and reference input (see Table 4.1 on page 197).
4. Know a few basic Rules for Plotting a Positive (180°) Root Locus (found in 5.2.1 & 5.2.2 Summary)
5. RULE 1. The n branches of the locus start at the poles of $L(s)$ and m branches end on the zeros of $L(s)$.
6. RULE 2. The loci are on the real axis to the left of an odd number of poles and zeros.
7. RULE 3*. The loci are attracted to zeros and repelled by poles. (Brother Smith's rule)
8. Understand what a root-locus plot represents and shows.
9. Understand the Ziegler-Nichols Tuning methods and what they provide.
10. Understand the Process Reaction Curve use in the Ziegler-Nichols tuning method and what it represents.
11. Understand how changing PID gains (K_p, K_i, K_d) affects percent overshoot, rise time, settling time, and steady-state error.
12. Know how to determine the finite and infinite zeros in a transfer function.
13. Know the form for a lead compensator and what it is used to achieve.
14. Know the form of a lag compensator and what it is used to achieve.
15. Know how to arrange characteristic equation in the right form to plot a root-locus.
16. Know how to select the best pole locations to achieve a dominate second order response.
17. Be able to identify locations on a root locus that produce a stable, marginally stable, and unstable response.
18. Be able to identify open-loop poles and zeros on a root-locus plot.
19. Be able to determine the transfer function and characteristic equation given a block diagram.
20. Know best practices for using pole-zero cancelation and for stabilizing a system with an unstable pole.
21. Know how to determine the steady-state constants: k_p, k_v, k_a. (see eq. 4.36 - 4.38).
22. Know the definition of a proper rational function.
23. Understand what a sensitivity function is, what it is used for, and what it tells you about a system.